

Measurement Units

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS We measure with two systems. **Customary** units include length (inch, foot, yard), weight (ounce, pound), and liquid volume (cup, pint, quart, gallon). **Metric** units include length (centimeter, meter, kilometer), mass (gram, kilogram), and liquid volume (milliliter, liter). To **convert within one system**, use the sizes: 12 inches = 1 foot, 3 feet = 1 yard, 16 ounces = 1 pound, 2 cups = 1 pint, 4 quarts = 1 gallon; 100 centimeters = 1 meter, 1,000 grams = 1 kilogram, 1,000 mL = 1 liter. Always **pick the unit that fits** the size of the object.

Round 1 — Retrieve It

1. Name the two systems we use to measure.

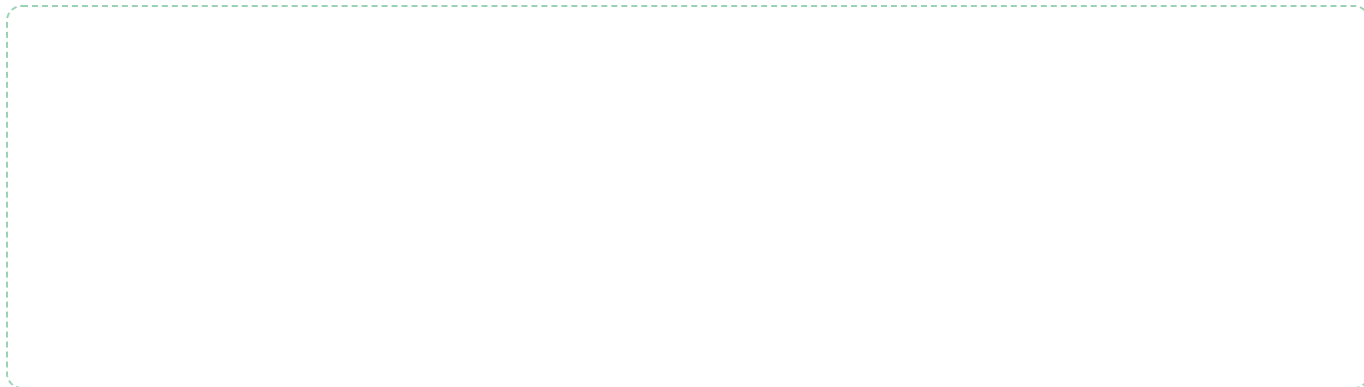
2. How many inches are in 1 foot?

3. Which metric unit would you use to measure the mass of an apple, gram or kilogram?

4. How many cups are in 1 pint?

Round 2 — Sketch It

Draw three boxes labeled length, weight, and liquid volume. In each box, list one customary unit and one metric unit that measures it.



Measurement Units — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. A rope is 2 yards long. How many feet is that? Show your thinking.

2. A bottle holds 2 liters. How many milliliters is that?

3. A jug holds 3 gallons. How many quarts is that?

Round 4 — Reason from Evidence

A friend says the length of a school bus is best measured in inches. Explain why that is not the best unit, and tell which unit fits better.

Measurement Units — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain how the customary system and the metric system are different. Give one unit from each.

C**Connect**

Connect measurement to something you do at home, such as cooking or measuring your height. Name the unit you would use.

E**Extend**

Pick the best unit for each: your pencil, a bag of dog food, and the water in a glass. Tell why each unit fits.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Measurement Units

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Customary and metric.
2. 12 inches.
3. Gram (an apple is light; kilograms are for heavier objects).
4. 2 cups.

Sketch It — look for

Length: inch or foot (customary) and centimeter or meter (metric). Weight: ounce or pound and gram or kilogram. Liquid volume: cup/quart/gallon and milliliter or liter.

Apply It

1. 6 feet ($2 \text{ yards} \times 3 \text{ feet per yard} = 6 \text{ feet}$).
2. 2,000 milliliters ($2 \times 1,000 \text{ mL}$).
3. 12 quarts ($3 \text{ gallons} \times 4 \text{ quarts per gallon}$).

Reason from Evidence — look for

Inches are very small, so a bus would be a huge number of inches. A larger unit such as feet or yards fits the size of a bus much better.

ACE — Extend look-for

Pencil: inches or centimeters (small length). Bag of dog food: pounds or kilograms (heavier weight). Water in a glass: ounces or milliliters (small liquid amount).

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.